



SEASONAL RAINFALL OUTLOOK FOR 2026/27 OVER SADC REGION

**SADC
CLIMATE SERVICES CENTRE (CSC)
and
MEMBER STATES**

26 – 28 August 2026



Climate drivers affecting Southern African rainfall

Summer rainfall in SADC region is driven by:

- large scale convergence associated with **ITCZ/CAB** (Congo Air Boundary)
- localized convection associated with **continental heating**
- convection associated with **meso-scale synoptic system** such as **tropical cyclones, tropical temperate troughs, cut-off lows**, etc.
- Interaction of **sub-tropical anti-cyclones, tropical convergence, mid-latitude westerlies and continental high-pressure systems**

These processes take place within **interacting air masses bringing moisture from tropical Atlantic Ocean and from tropical and sub-tropical Indian Ocean.**

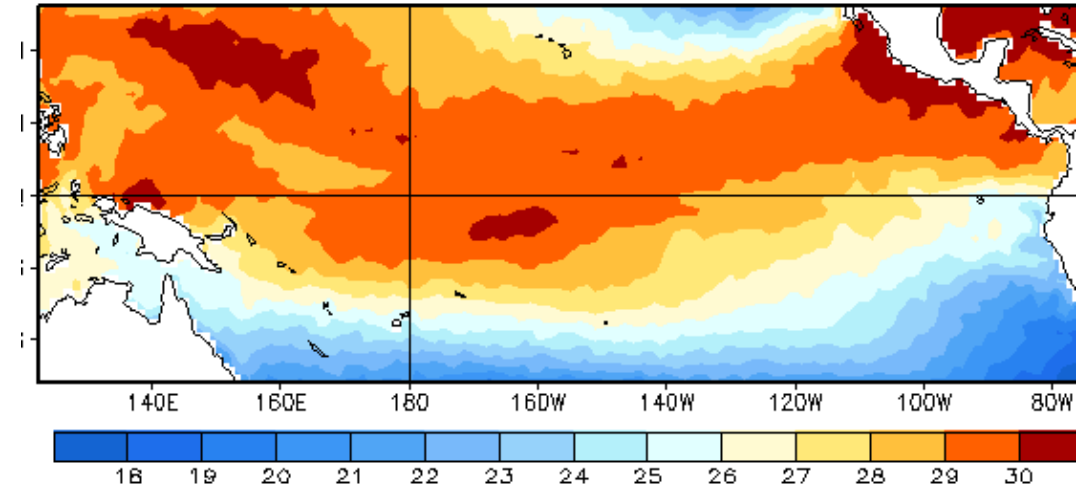
Large scale circulation dynamics are affected by teleconnections with ENSO, IOD (also SAM - but mostly in south-western tip of the continent).

For island states a different index - SIOD - is relevant.

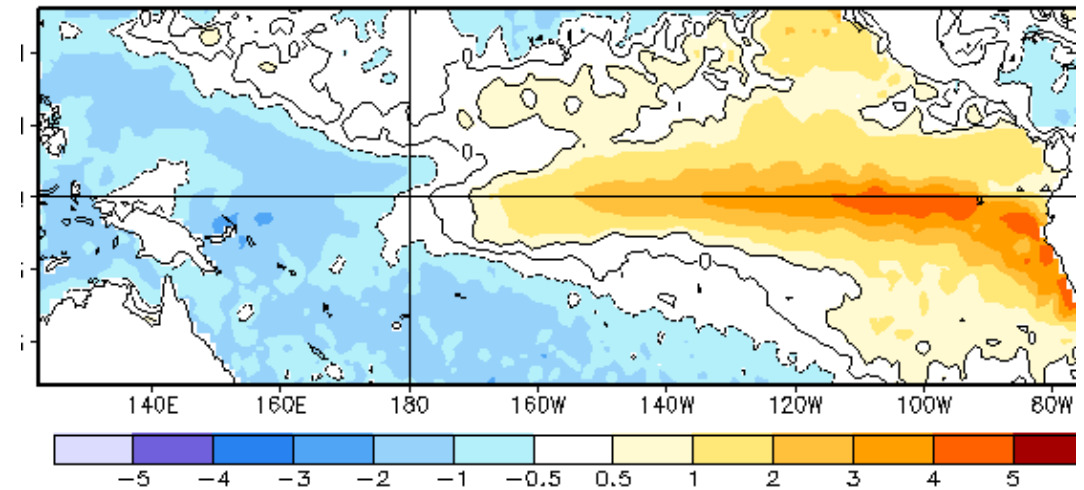


Current ENSO status

Observed Sea Surface Temperature (°C)



Observed Relative Sea Surface Temperature Anomalies (°C)



7-day Average Centered on 19 August 2026

- ENSO Status in Aug 2026:
 - Moderate **El Niño**
- Equatorial sea surface temperatures (SSTs) are **above** average across most of the equatorial Pacific Ocean.
- The tropical Pacific atmosphere is consistent with **El Niño**.



Performance of Climate Drivers – MAM and AMJ 2026

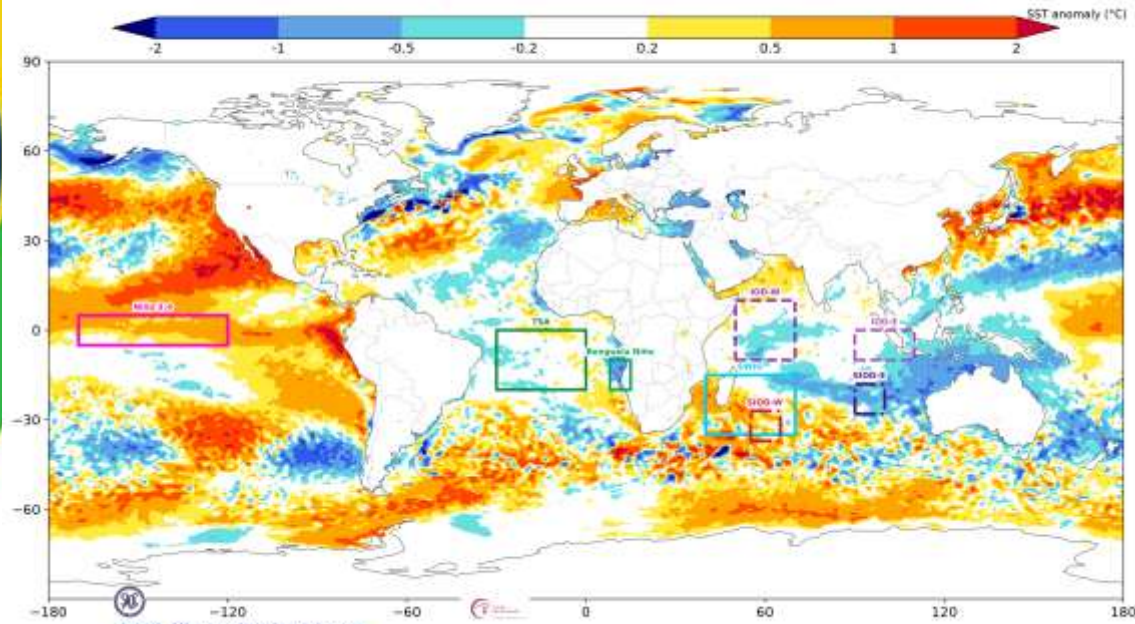


SADC CSC observed SST monitoring

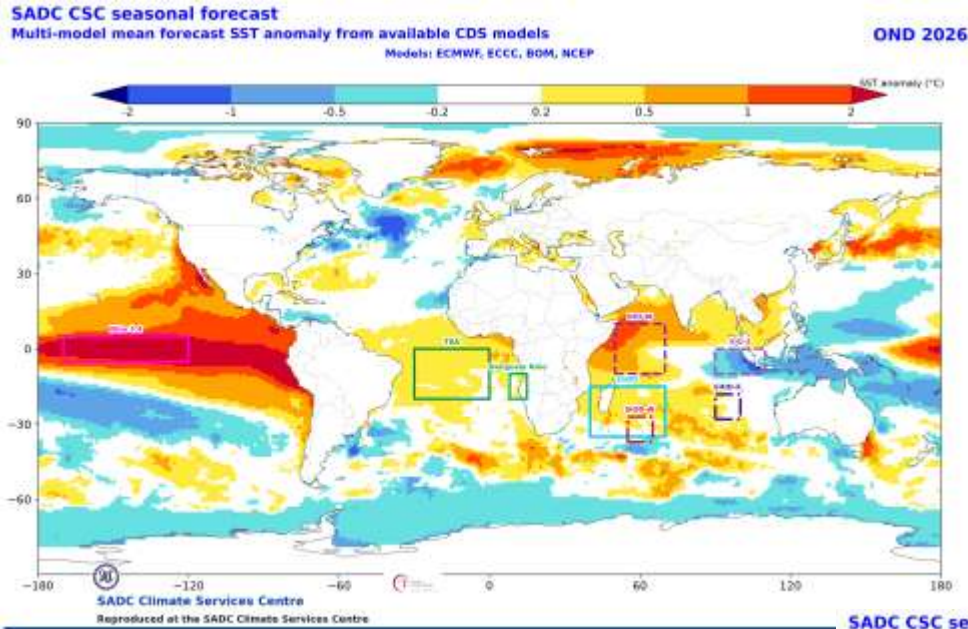
Observed 3-month mean SST anomaly relative to 1991-2020

Trend removed: linear monthly SST trend removed to reduce climate change effects (reference year: 2005.5)

MAM 2026

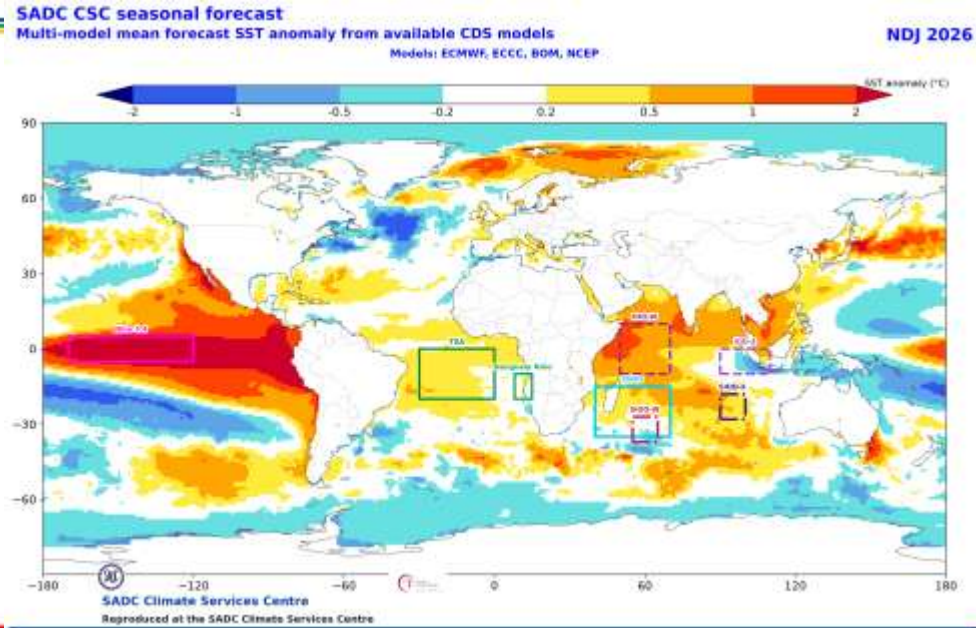


Projected State of Global Sea Surface Temperatures - Climate Drivers



Model: Multi-model
Period: OND 2026

ENSO State: **Strong El**
Nino: $>2^{\circ}\text{C}$



Model: Multi-model
Period: NDJ 2026

ENSO State: **Strong El**
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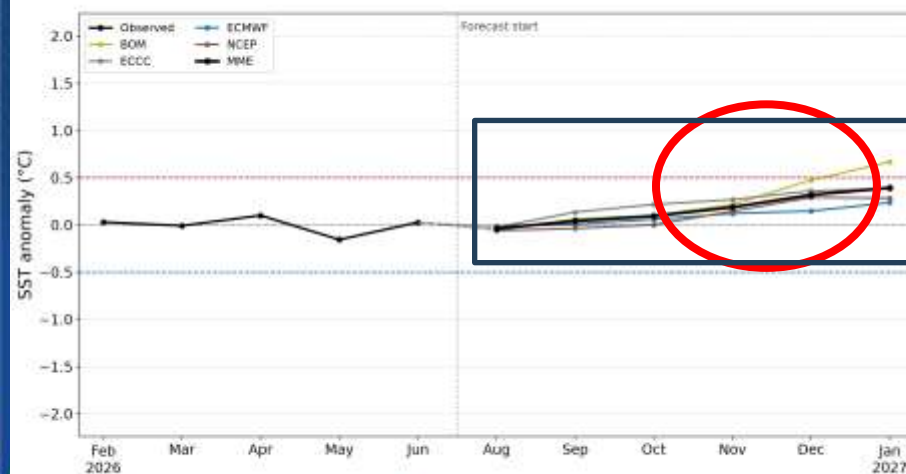


Projected State of Climate Drivers



Forecast of TSA
Aug2026 to Jan2027
Models: BOM, ECCO, ECMWF, NCEP

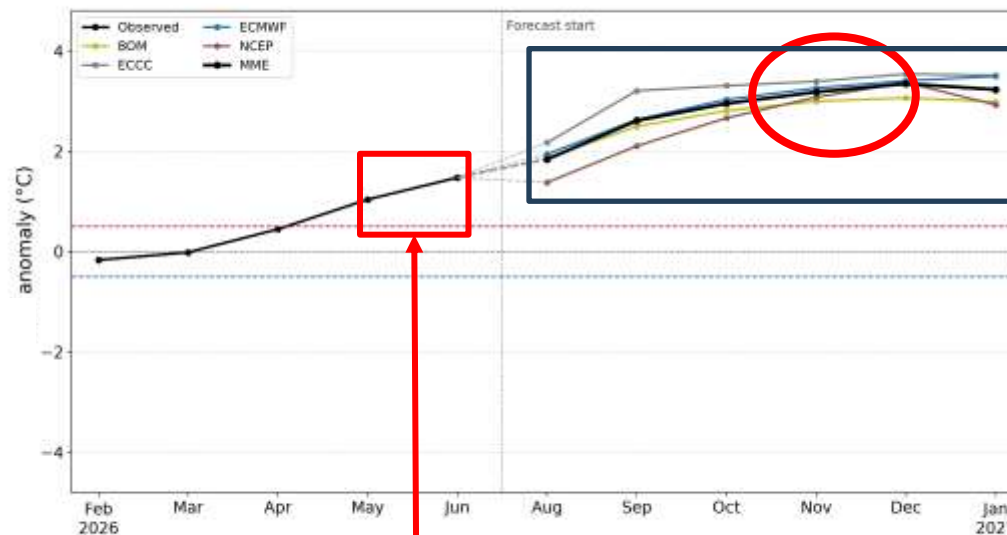
(Issued on Jul2026)



SADC Climate Services Centre
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Forecast of NINO34
Aug2026 to Jan2027
Models: BOM, ECCO, ECMWF, NCEP

(Issued on Jul2026)



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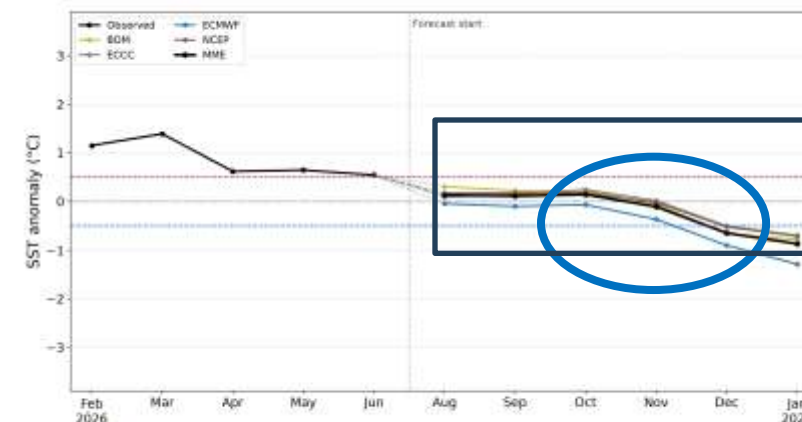
OND 2026: Between +3 and +4°C

Projected State: Strong El Nino

El Nino Status: Active
May and June 2026 Value:

Forecast of SIOD
Aug2026 to Jan2027
Models: BOM, ECCO, ECMWF, NCEP

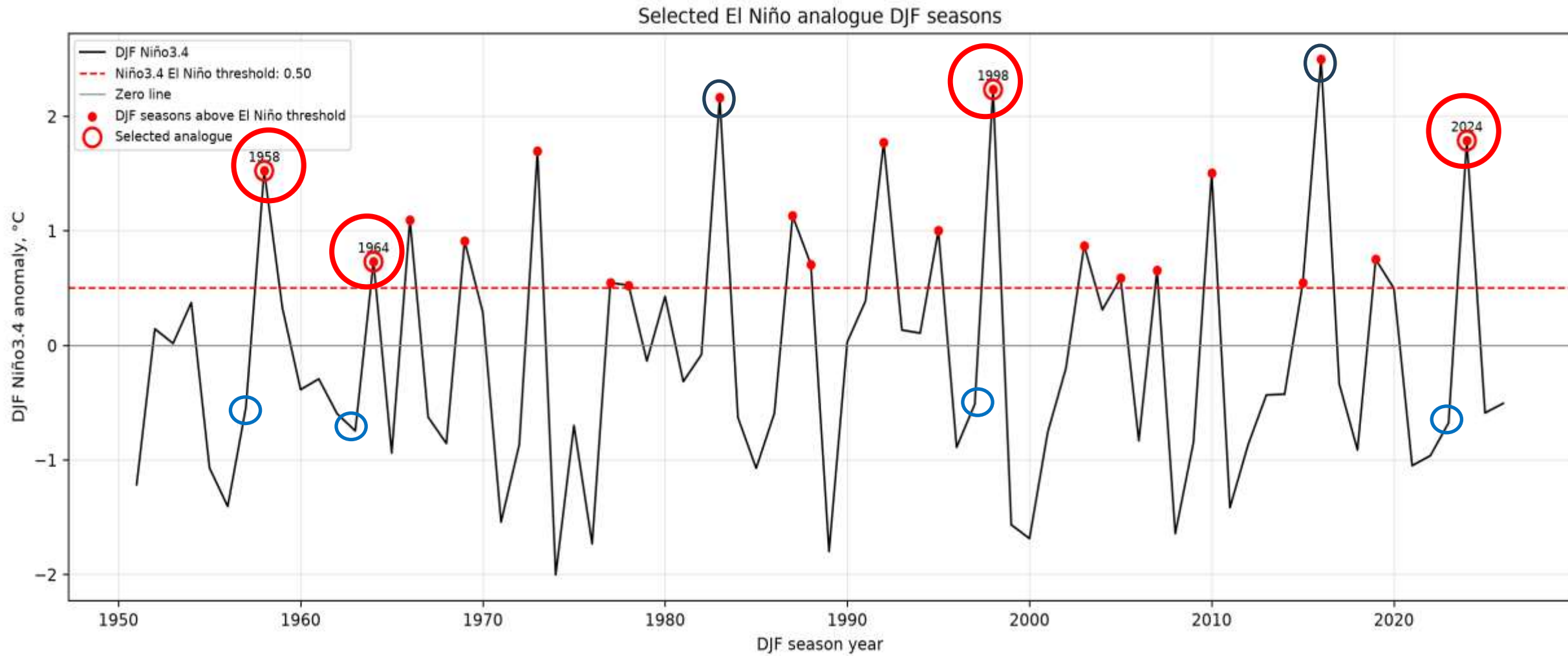
(Issued on Jul2026)



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Climate drivers analysis from past climate by Evolution



Climate drivers analysis from past climate by **Evolution**

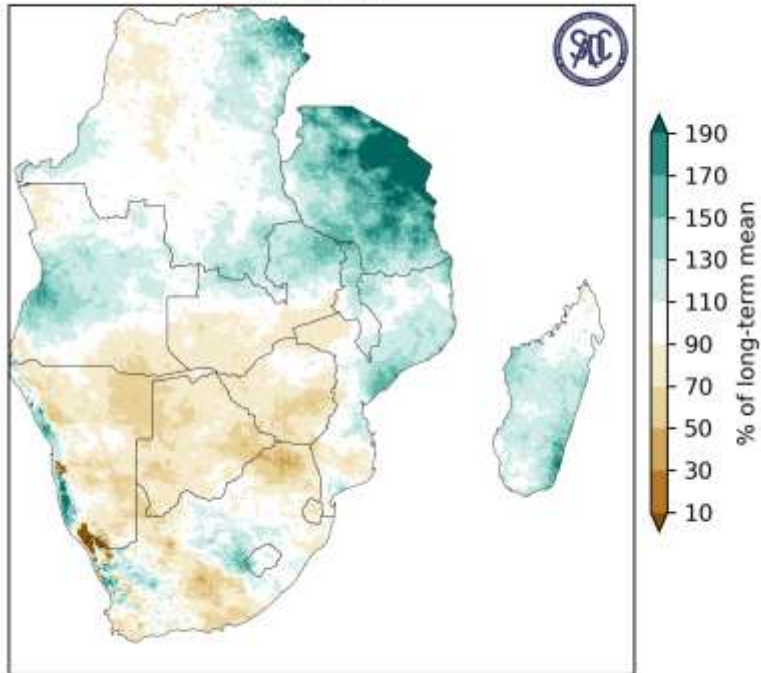


Climate Index / Metric	1997/98 Analog	2023/24 Analog	Projected 2026 Planning value
Niño 3.4 SST Anomaly	(Super EP) +2.4°C	(Strong EP) +2.0	Strong +2.5
Mean Austral Summer SOI	-2.1	-1.5	Predicted -1.3
Indian Ocean Dipole (DMI)	Extreme Positive +1.28	Extreme Positive +1.1	Developing Positive +0.6
SIOD Phase Status	Strongly Positive	Strongly Positive	Projected Positive Phase +0.5
TSA Index Anomaly	Critically Warm +0.7°C	Strongly Warm +0.8	Warm +0.5 to +0.7
TNA Index Anomaly	Near-Neutral -0.1°C	Critically Warm +1.2	Sustained Warm +0.9



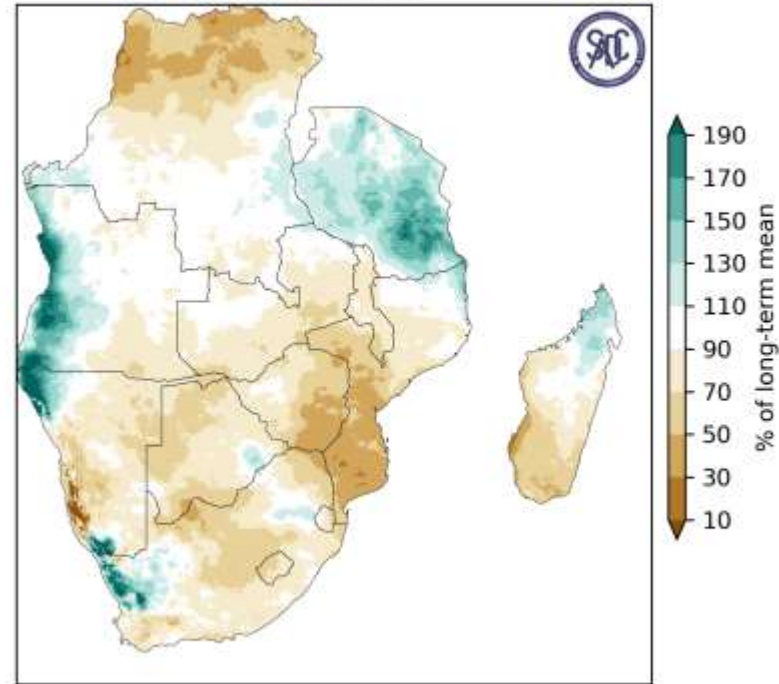
Rainfall performance in past similar ENSO states Analogues

Anomaly of total precipitation
DJF 1997-1998



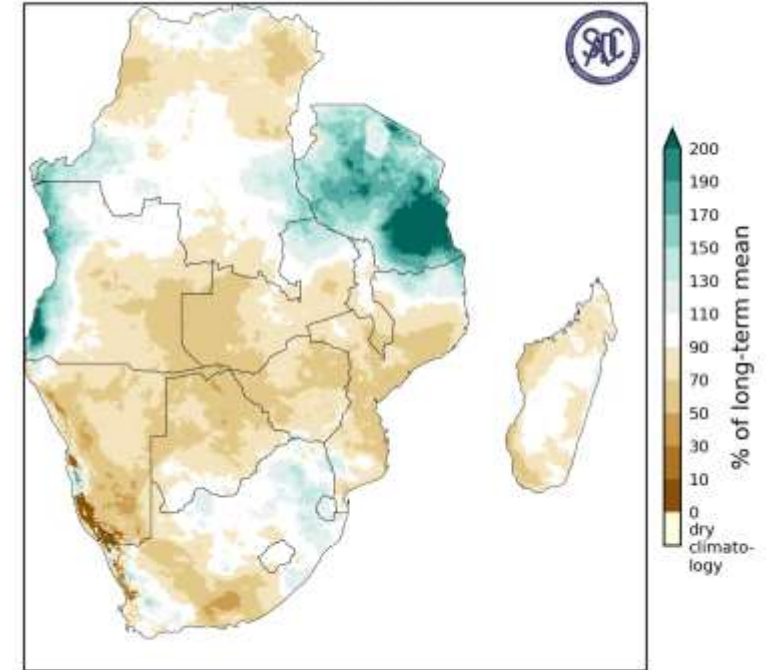
Data source: CHIRPS v2.0
Climatological period: 1991-2020

Anomaly of total precipitation
DJF 2015-2016



Data source: CHIRPS v2.0
Climatological period: 1991-2020

Anomaly of total precipitation
DJF 2023-2024



Data source: CHIRPS v2.0
Climatological period: 1991-2020
Regions where climatological rainfall is less than 1mm are masked out





(issued on Aug2026)

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Global Producing Centre's Climate Outlooks – NDJ / DJF 2026-27

Probabilistic Multi-Model Ensemble Forecast
 (Issued on Aug 2026)
Precipitation : DJF 2026

C3S multi-system seasonal forecast ECMWF/Met Office/Météo-France/CMCC/DWD/NCEP/JMA/ECCC/BOM
 Prob(precipitation > median)
 Nominal forecast start: 01/08/26
 Unweighted mean

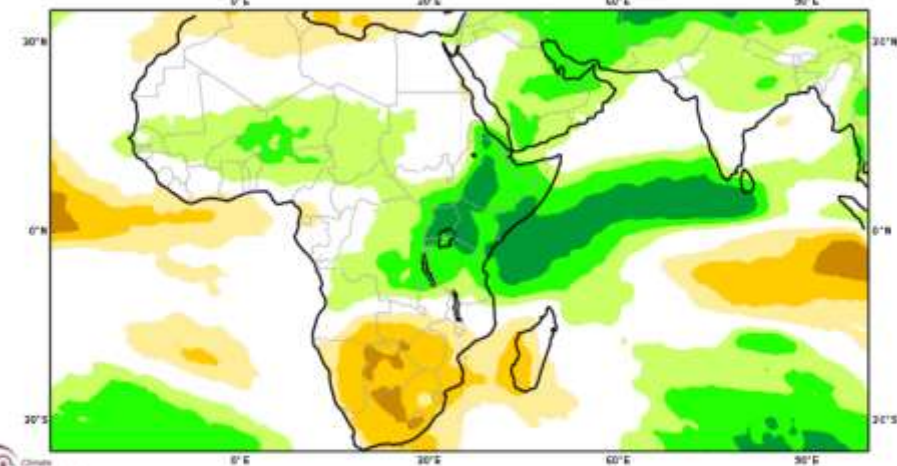
Probabilistic (tercile) forecast of rainfall for NDJ 2026
 issued in Aug 2026

Source: Multi-model ensemble (SEA551, CFSv2, CCSM4, GEOS525) calibrated at 5ADC CSC
 Skill evaluated by tercile probabilities against CHIRPS data over 1993-2015 period
 Values where Generalized ROC is less than 50 are masked out

Logos: Programme of the European Union, Copernicus, Climate Change Service, ECMWF



0-10% 10-20% 20-30% 30-40% 40-60% 60-70% 70-80% 80-90% 90-100%



Source: Multi-model ensemble (SEAS51, CFSv2, CCSM4, GEOS525) calibrated at SADC CSC
Skill evaluated by tercile probabilities against CHIRPS data over 1993-2015 period
Values where Generalized ROC is less than 50 are masked out





Part 2 – Seasonal Outlook

Ms. Surekha Ramessur
Senior Program Officer
Meteorology Unit

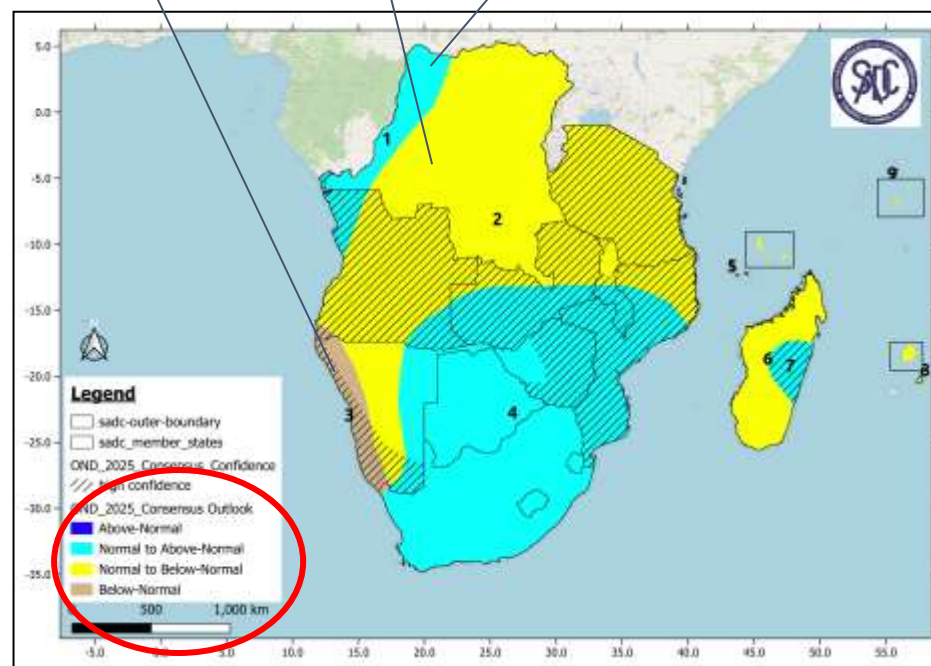
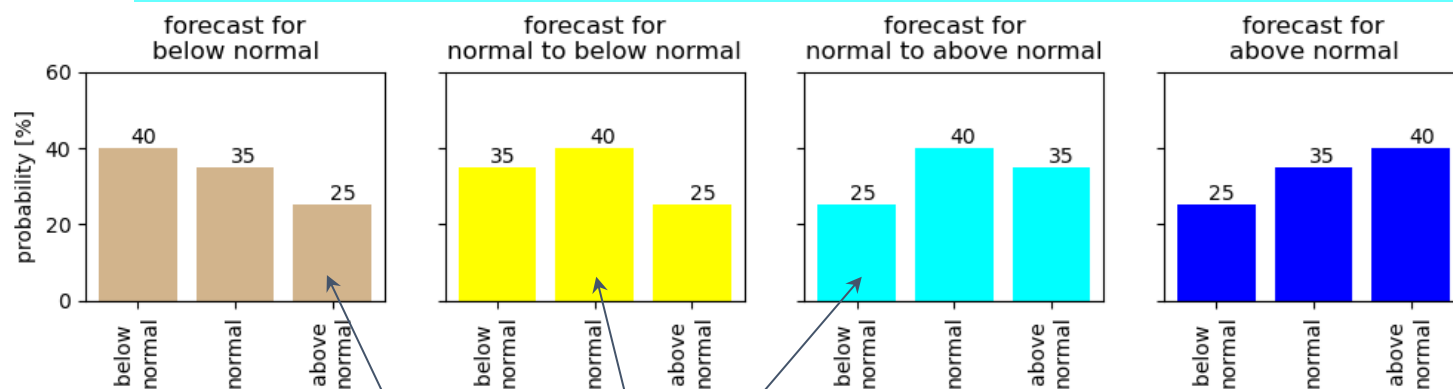


Background information

- The Climate experts meeting was conducted from 17 - 24 August 2026
- Using statistical analysis, other climate prediction schemes and expert interpretation, **likelihoods of above-normal, normal-to-above, normal-to-below and below-normal rainfall** for each zone have been forecasted for overlapping three-monthly periods
- Outlook is relevant for large areas and applicable at regional scale only. **National or local area climate outlooks are available from the National Meteorological and Hydrological Services (NMHSs) of the SADC region.**



What does SARCOF forecast map mean?



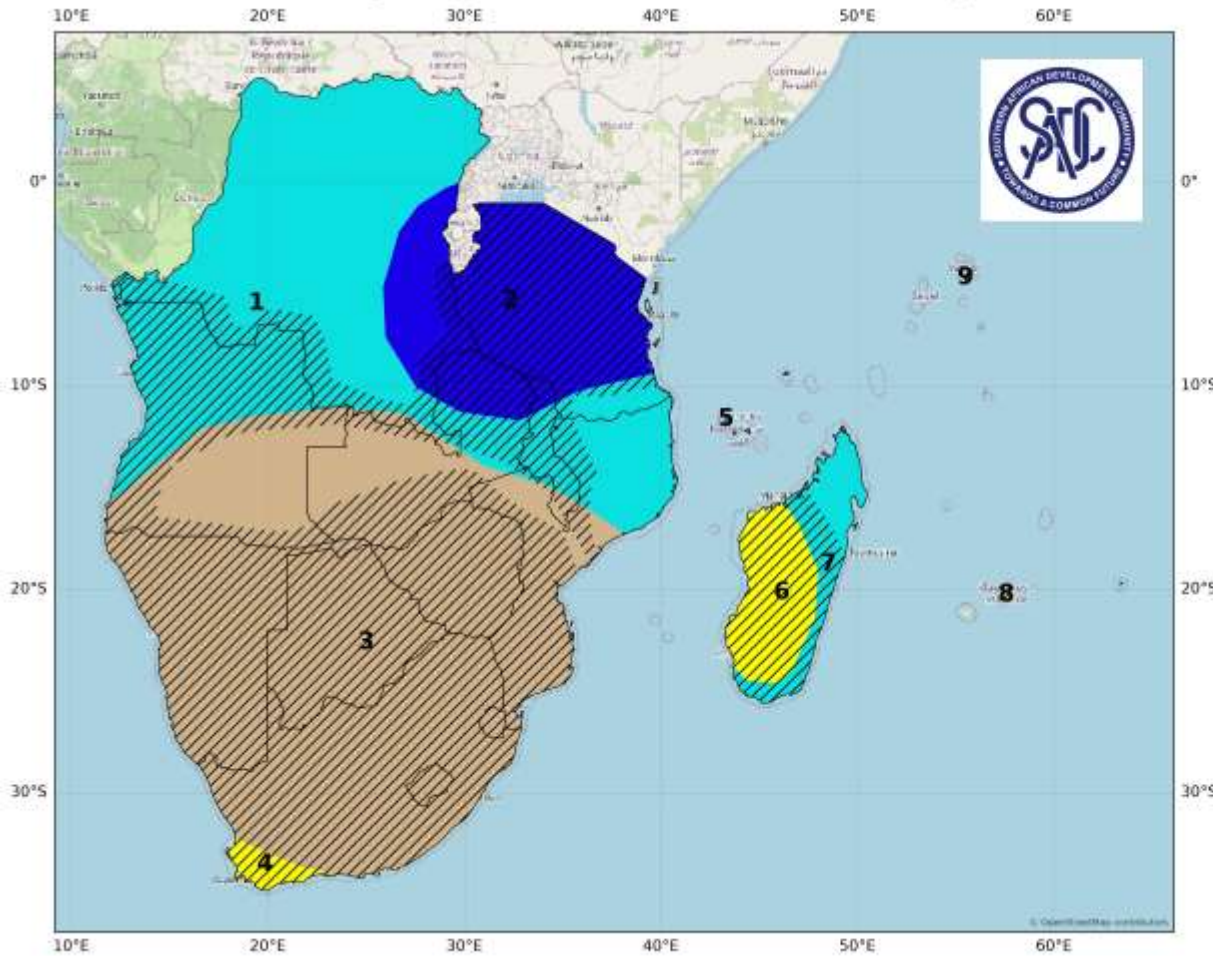
Forecast for below normal **does not tell us** that rainfall **will be** below normal.

It tells us that there is an **increased chance** of below normal rainfall.

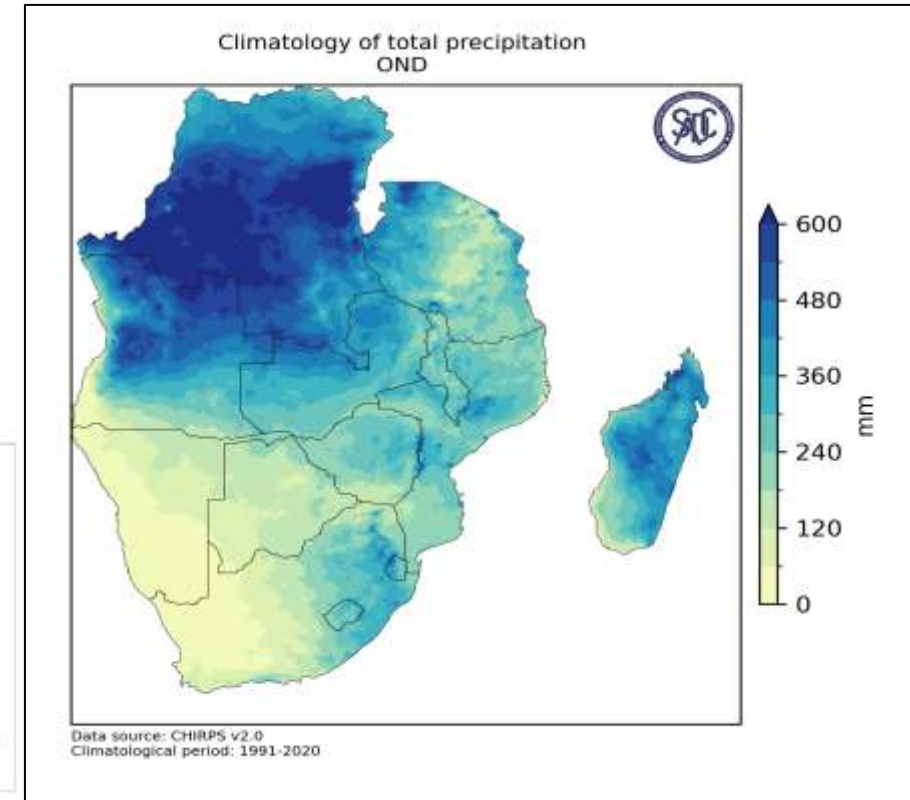


Seasonal Outlook for OND 2026

OND Merged Regional Forecast with Confidence Overlay

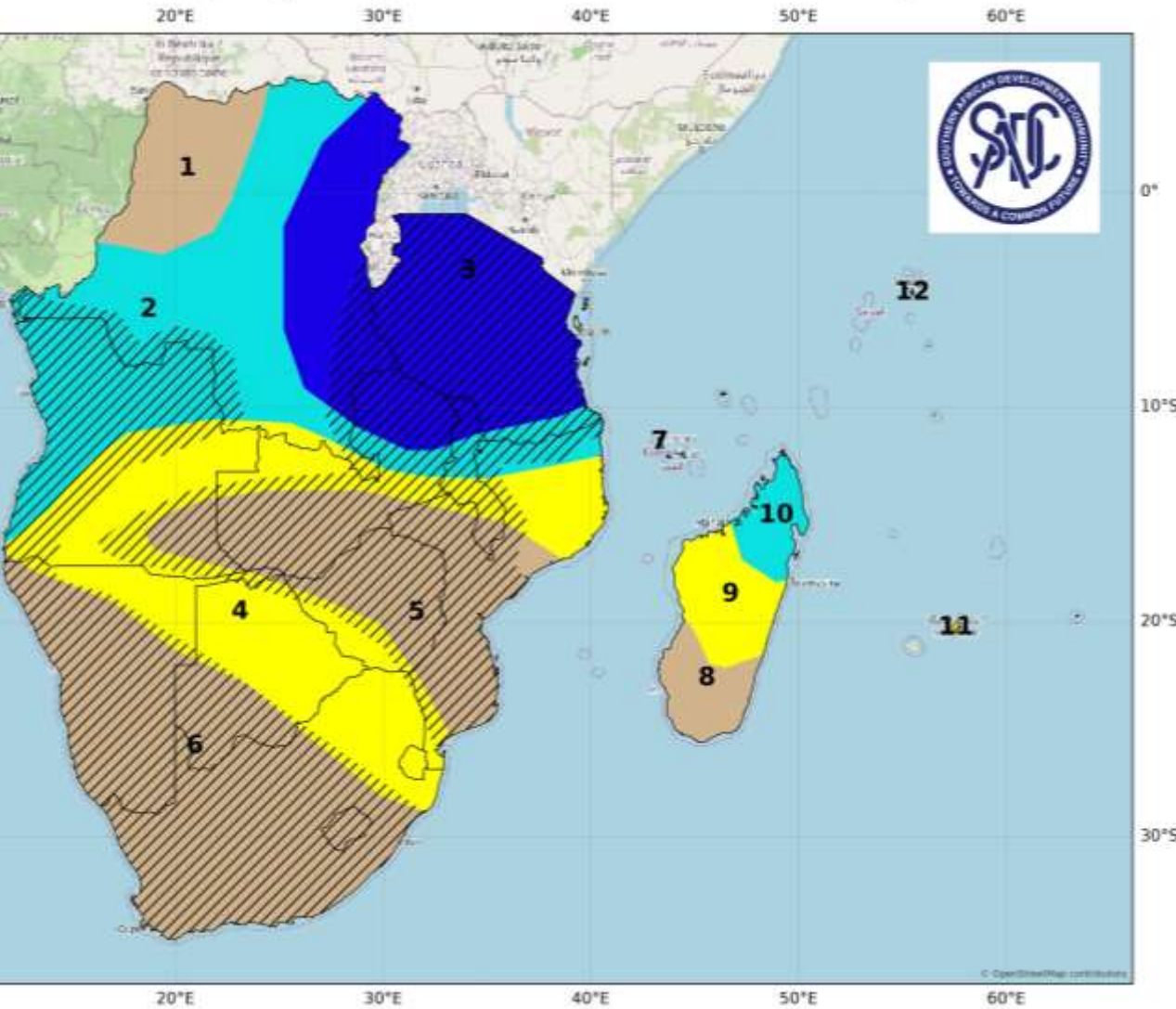


OND Long-term Rainfall



Seasonal Outlook for NDJ 2026/27

NDJ Merged Regional Forecast with Confidence Overlay



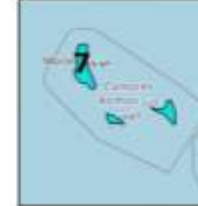
Seychelles



Mauritius



Comoros



Legend

- High confidence
- Below-Normal
- Normal to Below-Normal
- Normal to Above-Normal
- Above-Normal

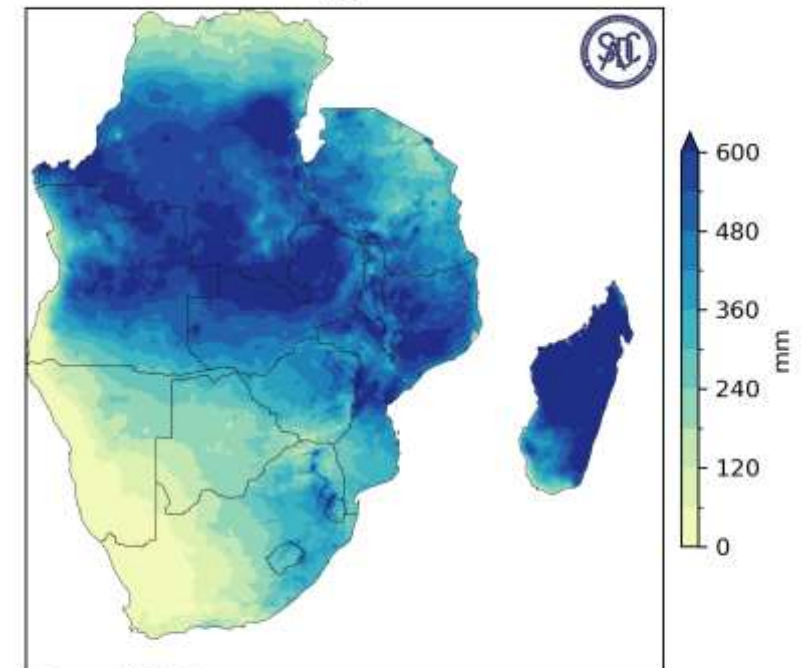
N

Scale



NDJ Long-term Rainfall

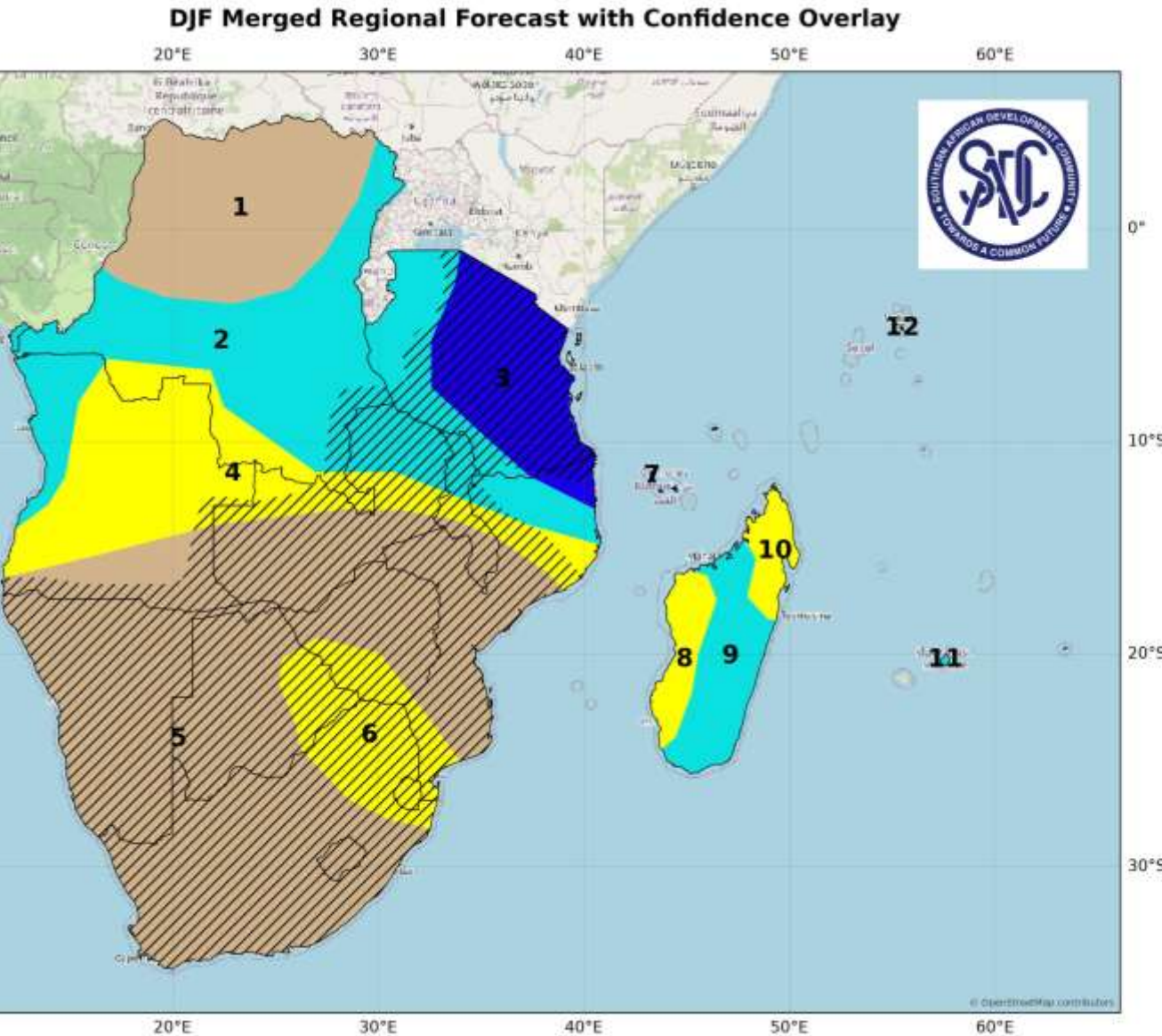
Climatology of total precipitation NDJ



Data source: CHIRPS v2.0
Climatological period: 1991-2020



Seasonal Outlook for DJF 2026/27



Legend

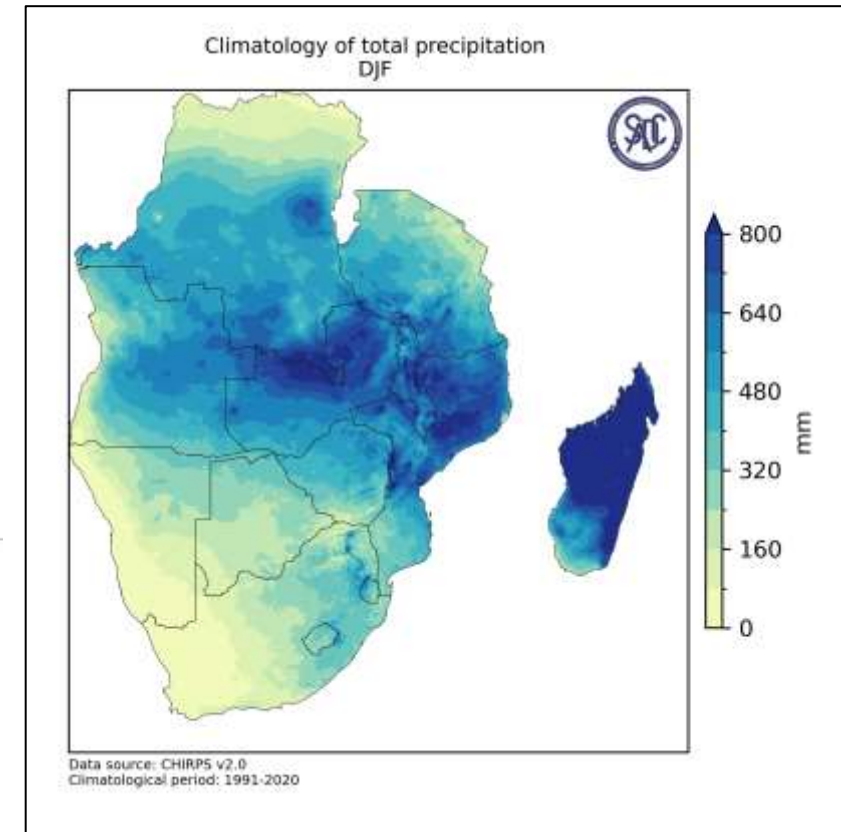
- High confidence
- Below-Normal
- Normal to Below-Normal
- Normal to Above-Normal
- Above-Normal



Scale

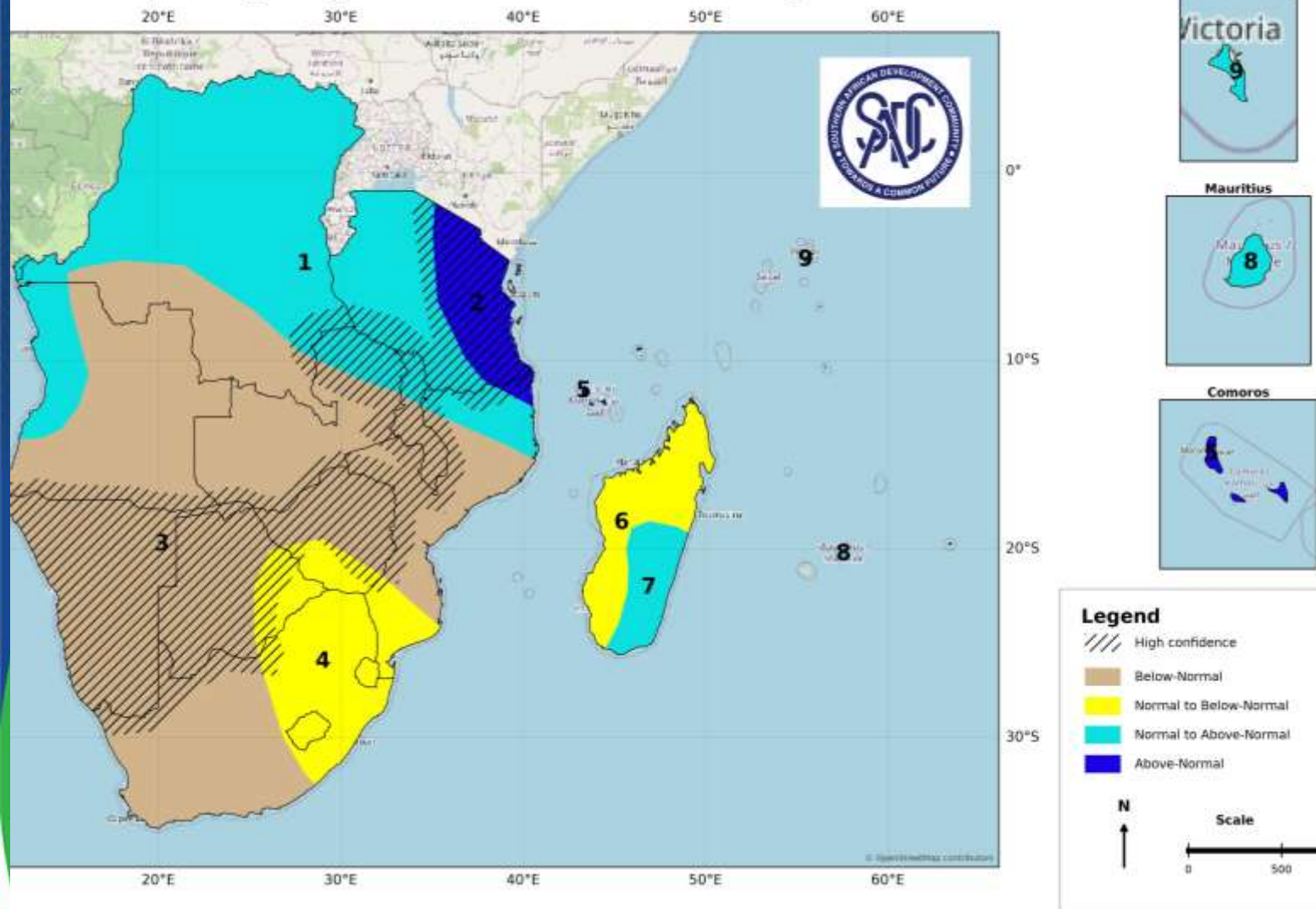


DJF Long-term Rainfall

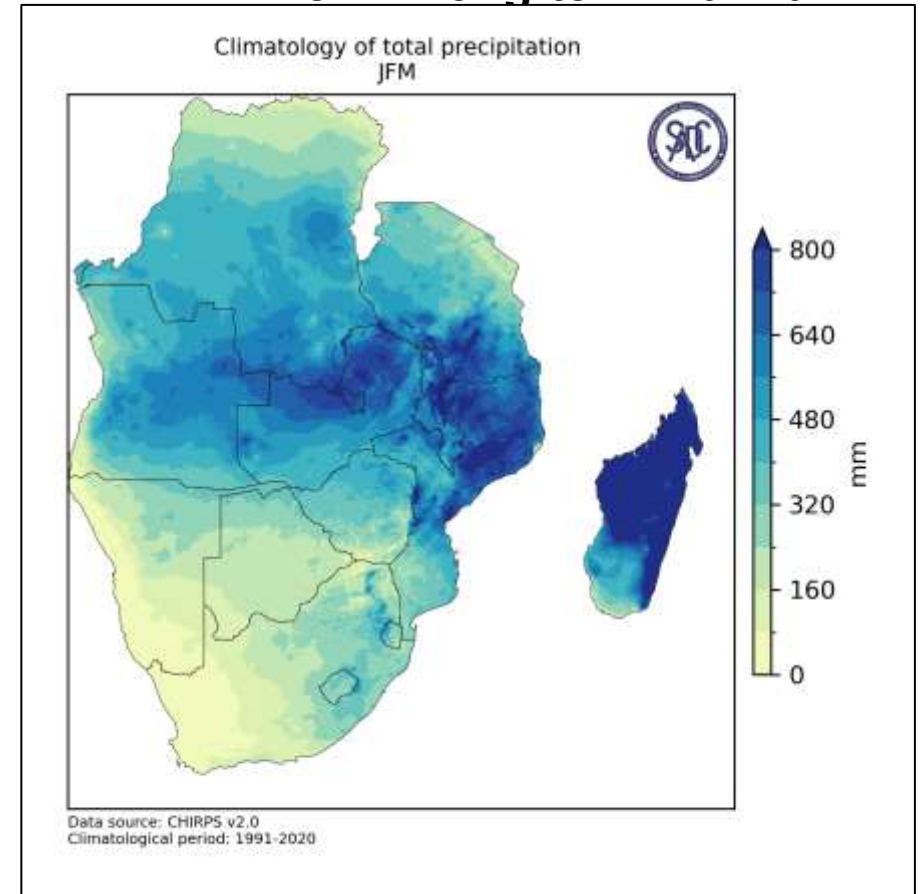


Seasonal Outlook for JFM 2027

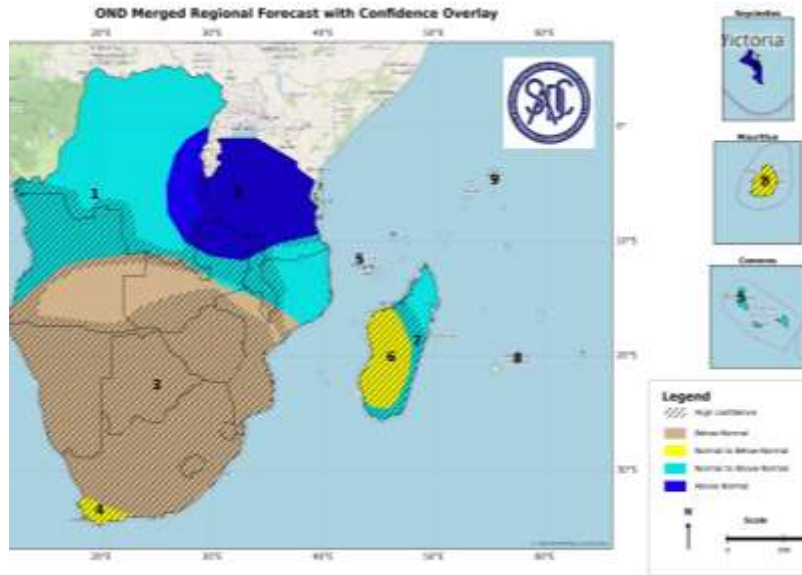
JFM Merged Regional Forecast with Confidence Overlay



JFM Long-term Rainfall

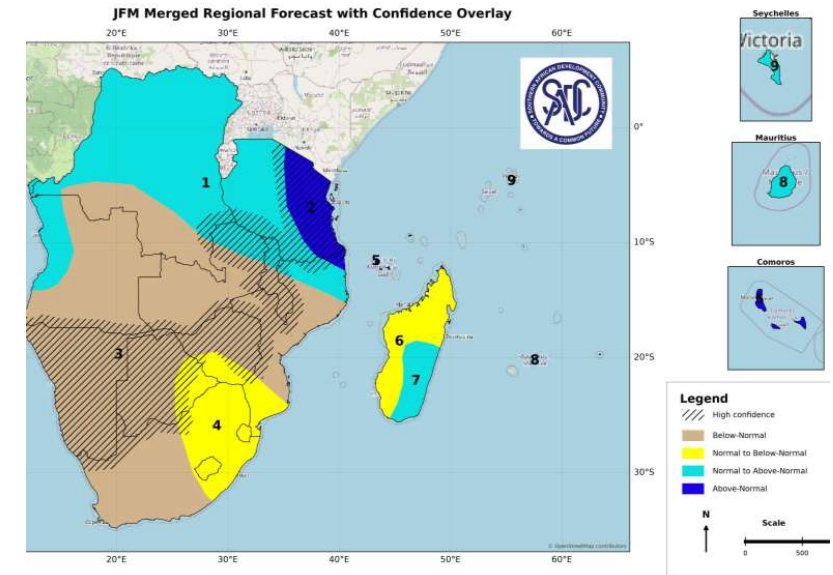


Summary of the Outlook



OND 2026

- **Below normal** rains are expected in Much of Angola, southern Zambia, Zimbabwe, Mozambique, Namibia, Botswana, most of South Africa, Eswatini and Lesotho.
- **Normal to below-normal** are expected in Southwestern coastal South Africa, western and southwestern Madagascar and Mauritius.
- **Normal to above-normal** are expected in most of the Democratic Republic of Congo (DRC), northwestern Angola, northern Zambia, central Malawi and northern Mozambique, as well as the southern fringes of Tanzania, Comoros, North and Eastern Madagascar
- **Above-normal** are expected in Eastern DRC, most of Tanzania, the extreme northern part of Malawi, extreme northern part of Zambia and Seychelles



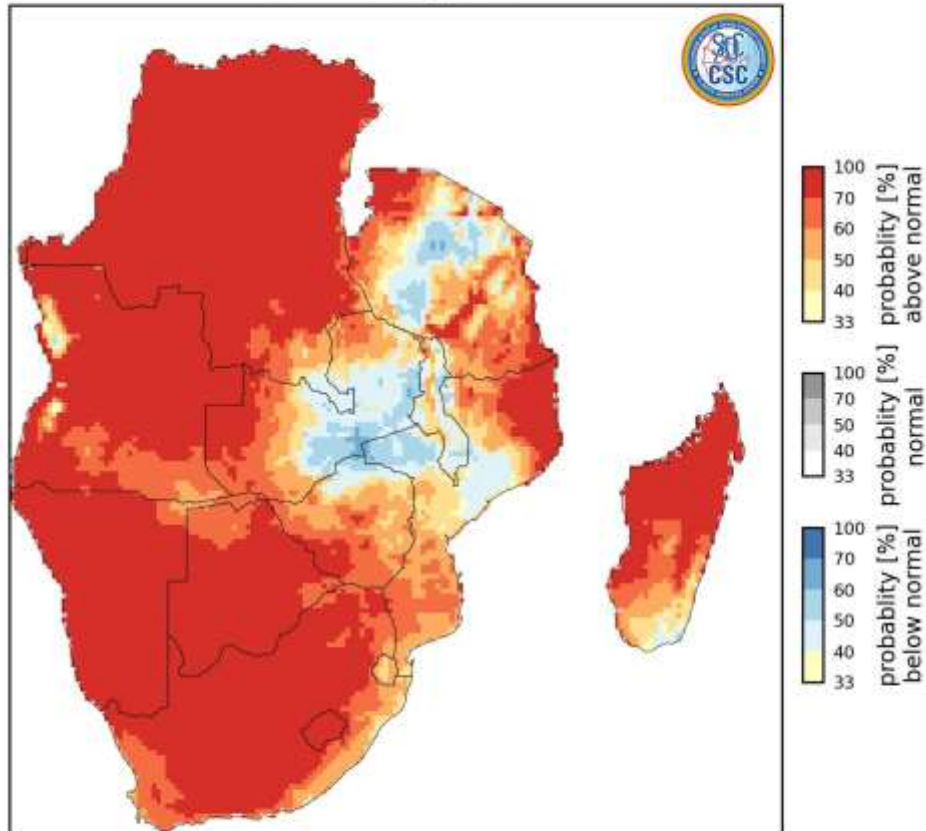
JFM 2027

- **Below-normal** are expected in the southern DRC, bulk of Angola, bulk of Zambia, central and southern Malawi, central Mozambique, Namibia and bulk of Botswana, the western half of South Africa, and bulk of Zimbabwe.
- **Normal to below-normal** are expected in the eastern half of South Africa, Lesotho, Eswatini, southern Zimbabwe, extreme southern Mozambique and north and western Madagascar
- **Normal to above-normal** are expected in the most of the Democratic Republic of Congo (DRC), extreme northwestern Angola, extreme northern Zambia, northern Malawi, northern parts Mozambique, western most of Tanzania, southeastern Madagascar, Mauritius, Seychelles.
- **Above normal rains** are expected in eastern Tanzania, extreme northern tip of Mozambique and Comoros.

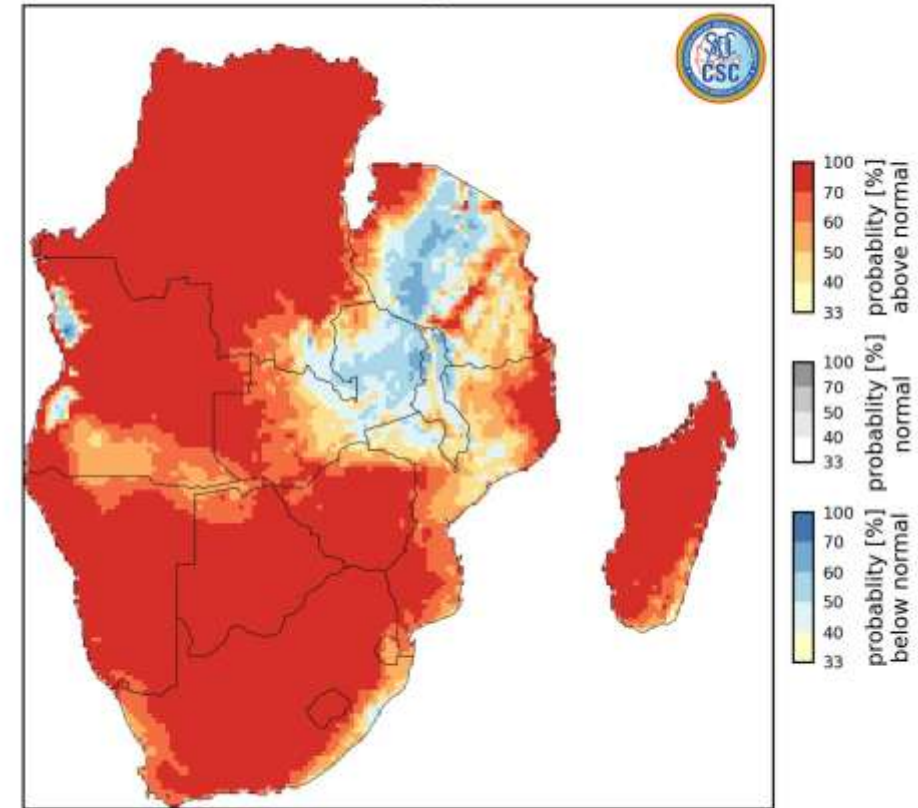


Temperature outlook for OND 2026 & NDJ 2026/27

Probabilistic (tercile) forecast
of mean temperature for OND 2026
issued in Aug 2026



Probabilistic (tercile) forecast
of mean temperature for NDJ 2026
issued in Aug 2026



For Your Attention



Thank You Merci Obbrigado Asante



@sadccsc



@sadccsc8480



<http://csc.sadc.int/en/>

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